

Table of Contents

Module 01: Introduction to Ethical Hacking	01
Information Security Overview	04
Hacking Concepts and Hacker Classes	18
Ethical Hacking Concepts	26
Hacking Methodologies and Frameworks	45
Information Security Controls	67
Information Security Laws and Standards	98
 Module 02: Footprinting and Reconnaissance	 121
Footprinting Concepts	124
Footprinting through Search Engines	131
Footprinting through Internet Research Services	157
Footprinting through Social Networking Sites	197
Whois Footprinting	212
DNS Footprinting	219
Network and Email Footprinting	230
Footprinting through Social Engineering	247
Footprinting Tasks using Advanced Tools and AI	254
Footprinting Countermeasures	275
 Module 03: Scanning Networks	 279
Network Scanning Concepts	282
Host Discovery	300
Port and Service Discovery	318
OS Discovery	357
Scanning Beyond IDS and Firewall	378
Network Scanning Countermeasures	408
 Module 04: Enumeration	 423
Enumeration Concepts	426
NetBIOS Enumeration	434
SNMP and LDAP Enumeration	447
NTP and NFS Enumeration	467
SMTP and DNS Enumeration	478

Other Enumeration Techniques	511
Enumeration Countermeasures	534
Module 05: Vulnerability Analysis	543
Vulnerability Assessment Concepts	547
Vulnerability Assessment Tools	581
Vulnerability Assessment Reports	614
Module 06: System Hacking	621
Gaining Access	624
Escalating Privileges	762
Maintaining Access	824
Clearing Logs	945
Module 07: Malware Threats	997
Malware and APT Concepts	1000
Fileless Malware Concepts	1093
AI-based Malware Concepts	1111
Malware Analysis	1126
Malware Countermeasures	1245
Module 08: Sniffing	1265
Sniffing Concepts	1268
Sniffing Techniques	1284
Sniffing Tools	1356
Sniffing Countermeasures	1365
Module 09: Social Engineering	1377
Social Engineering Concepts	1380
Human-based Social Engineering Techniques	1389
Computer-based Social Engineering Techniques	1401
Mobile-based Social Engineering Techniques	1436
Social Engineering Countermeasures	1444
Module 10: Denial-of-Service	1463
DoS/DDoS Concepts	1466
DoS/DDoS Attack Techniques	1484

DoS/DDoS Attack Countermeasures	1515
Module 11: Session Hijacking	1543
Session Hijacking Concepts	1546
Application-Level Session Hijacking	1558
Network-Level Session Hijacking	1581
Session Hijacking Countermeasures	1597
Module 12: Evading IDS, Firewalls, and Honeypots	1621
IDS, IPS, and Firewall Concepts	1624
IDS, IPS, and Firewall Solutions	1653
Evading IDS/Firewalls	1667
Evading NAC and Endpoint Security	1723
Honeypot Concepts and Detecting Honeypots	1763
IDS/Firewall Evasion Countermeasures	1777
Module 13: Hacking Web Servers	1787
Web Server Concepts	1790
Web Server Attacks	1812
Web Server Attack Methodology	1838
Web Server Attack Countermeasures	1885
Module 14: Hacking Web Applications	1917
Web Application Concepts	1921
Web Application Threats	1930
Web Application Hacking Methodology	2017
Web API and Webhooks	2154
Web Application Security	2205
Module 15: SQL Injection	2285
SQL Injection Concepts	2288
Types of SQL Injection	2297
SQL Injection Methodology	2311
Evasion Techniques	2400
SQL Injection Countermeasures	2416

Module 16: Hacking Wireless Networks	2439
Wireless Concepts	2442
Wireless Encryption	2456
Wireless Threats	2475
Wireless Hacking Methodology	2486
Wireless Attack Countermeasures	2562
Module 17: Hacking Mobile Platforms	2579
Mobile Platform Attack Vectors	2582
Hacking Android OS	2620
Hacking iOS	2682
Mobile Device Management	2719
Mobile Security Guidelines and Tools	2732
Module 18: IoT and OT Hacking	2765
IoT Concepts and Attacks	2770
IoT Hacking Methodology	2830
IoT Attack Countermeasures	2883
OT Concepts and Attacks	2905
OT Hacking Methodology	2964
OT Attack Countermeasures	3003
Module 19: Cloud Computing	3023
Cloud Computing Concepts	3027
Cloud Computing Threats	3083
Cloud Hacking	3148
AWS Hacking	3170
Microsoft Azure Hacking	3244
Google Cloud Hacking	3275
Container Hacking	3292
Cloud Security	3310
Module 20: Cryptography	3365
Cryptography Concepts and Encryption Algorithms	3368
Applications of Cryptography	3414
Cryptanalysis Methods and Cryptography Attacks	3462
Cryptography Attack Countermeasures	3503